

UCLA Math 131A, Winter 2025
Midterm 1 Practice

Instructor: Linfeng Li

Monday, January 29

Please print your name and student ID below.

Name: _____

Student ID: _____

Please read all of the following rules before proceeding.

- The exam worth 40 points in total.
- Show all your work, and precisely state any theorems that you use.
- You are not allowed to use calculators or notes during the test, but you may use one-page, double-sided, and handwritten cheat sheet.
- Good luck!

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
Total:	40	

1. (10 points) Circle TRUE or FALSE for each of the following statement. No justification is needed.

- | | | | |
|-------|------|-------|--|
| (i) | True | False | The negation of the statement " $P \implies Q$ " is " P and $\neg Q$ ". |
| (ii) | True | False | The set $\{a \in \mathbb{Q} : a < 1\}$ attains both maximum and supremum. |
| (iii) | True | False | If $S \subseteq \mathbb{R}$ is a set with finitely many elements, then S has both maximum and minimum. |
| (iv) | True | False | Let $S \subseteq \mathbb{R}$ be nonempty. If the maximum of S exists, then the maximum is also the supremum of S . |
| (v) | True | False | The negation of the assertion " $\exists r \in \mathbb{R}, \text{ s.t. } \forall x \in [a, b], f(x) < r$ " is " $\forall r \in \mathbb{R}, \exists x \in [a, b], \text{ s.t. } f(x) < r$ " |

2. (10 points) Use induction to prove the sum-of-squares formula

$$1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

3. (a) (5 points) Let r be a nonzero rational number and x be an irrational number. Prove that rx is irrational.

- (b) (5 points) Let $a, b \in \mathbb{R}$. Prove that if $a \leq b + \epsilon$ for any $\epsilon > 0$, then $a \leq b$.

4. (a) (5 points) Let E, F be two nonempty bounded subsets of $\mathbb{R}$. Prove that

$$\sup(E \cup F) \geq \max\{\sup(E), \sup(F)\}.$$

- (b) (5 points) Prove that the greatest lower bound of the set $\{\frac{1}{n} : n \in \mathbb{N}\}$ is 0.